Heap File Organization and Free-Space Management

Heap File Organization

Definition

Heap file organization is the simplest method of storing records in a file. Records can be placed **anywhere** within the file where there is free space.

Characteristics

- Records are stored in **unordered fashion**.
- New records are placed in the first block with available free space.
- Records usually **do not move once allocated**, unless space reorganization or compaction is performed.
- Efficient mechanisms are required to quickly find blocks that have free space.

Advantages

- Simple to implement.
- Fast insertion (record can be placed wherever space is found).
- Suitable for small databases or when queries are not dependent on record order.

Disadvantages

- Searching is slower since records are unsorted.
- Range queries are inefficient.
- May lead to **fragmentation** and wasted space.

Free-Space Management

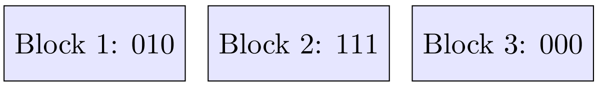
To efficiently manage free space within a heap file, a structure called a **Free-Space Map (FSM)** is maintained.

Free-Space Map (FSM)

- A lightweight data structure to keep track of how much space is free in each block.
- Typically implemented as an **array**, where each entry corresponds to a block of the heap file.
- Each entry is **a few bits to a byte**, storing an approximation of how much space is available.
- Example: If 3 bits are used per block, then each entry can store values 0–7.
- Value divided by 8 indicates the fraction of the block that is free.

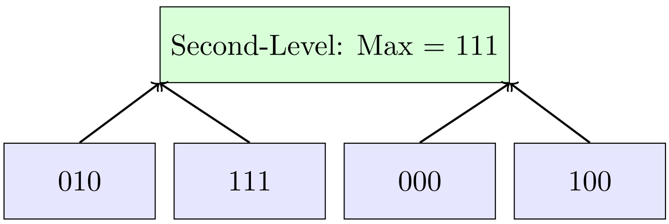
Example: First-Level Free-Space Map

$$\text{Block 1: } 010 \quad (2/8 = 25\% \text{ free})$$
$$\text{Block 2: } 111 \quad (7/8 \approx 87\% \text{ free})$$
$$\text{Block 3: } 000 \quad (0\% \text{ free})$$



Second-Level Free-Space Map

- To make searching faster, a hierarchy of free-space maps can be used.
- Each entry in the second-level FSM stores the **maximum free-space value** among a group of blocks in the first-level FSM.
- This allows quick identification of blocks that are more likely to have space.



Properties of FSM

- **Periodically written to disk** to maintain persistence.
- Slightly **outdated values** are acceptable, since discrepancies are fixed during insertion/deletion.
- Helps avoid scanning the entire file for free space, reducing overhead.

Summary

Key Points

- Heap file organization is simple and allows fast insertions, but has slower searches.
- Free-space maps enable efficient management of block space utilization.
- A second-level FSM improves performance by summarizing space availability across multiple blocks.
- Widely used in modern DBMS (e.g., PostgreSQL).